
```

clc;
clear;
close all;

% Input Signal
t = 0:0.001:1;
x = sin(2*pi*5*t);

% -----
% Uniform Quantization - 8 Levels
% -----
L = 8;

x_uni = round((x + 1) * (L - 1) / 2);
x_uni = (x_uni * 2 / (L - 1)) - 1;

% -----
% Non-Uniform Quantization - Mu Law
% -----
mu = 255;

% Compression
x_comp = sign(x) .* ...
    log(1 + mu * abs(x)) / log(1 + mu);

% Quantization
x_non = round((x_comp + 1) * (L - 1) / 2);

% Convert back to normalized range
x_non = (x_non * 2 / (L - 1)) - 1;

% -----
% Plot Results
% -----
figure('Name', 'Uniform and Non-Uniform Quantization', ...
    'NumberTitle', 'off');

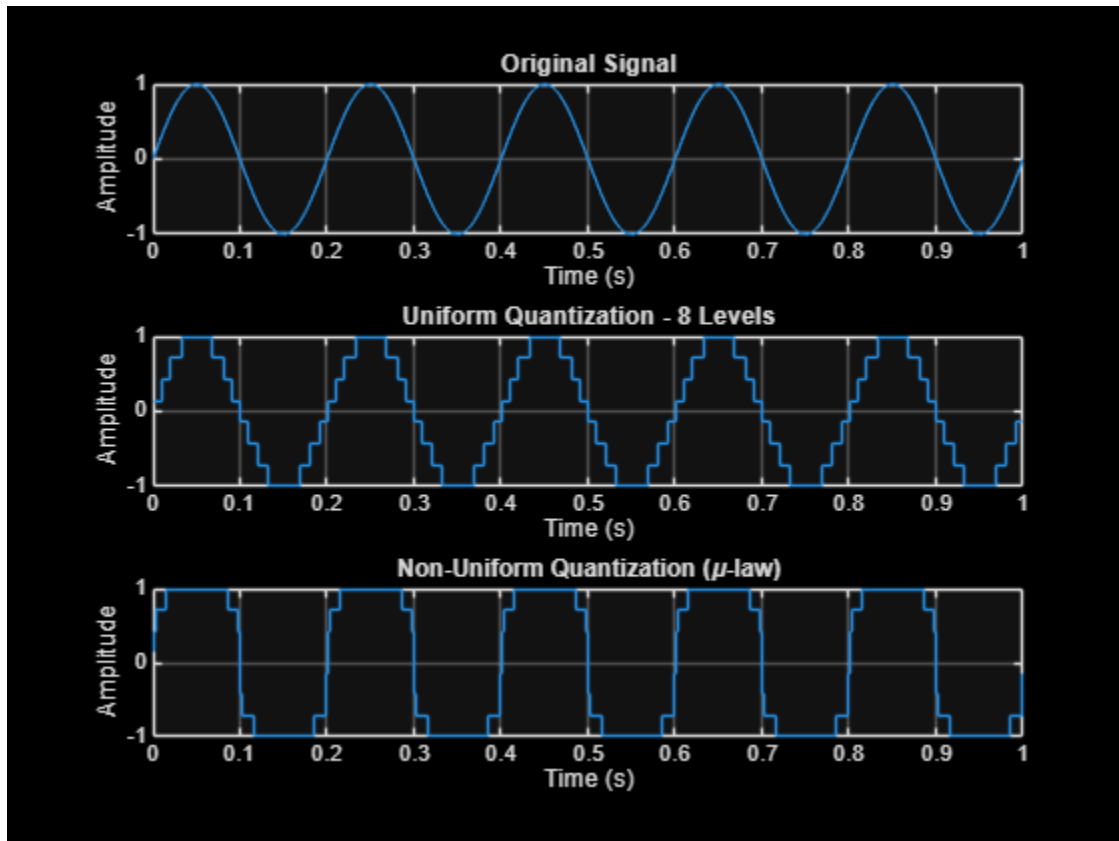
subplot(3,1,1);
plot(t,x);
title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
stairs(t,x_uni);
title('Uniform Quantization - 8 Levels');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(3,1,3);

```

```
stairs(t,x_non);  
title('Non-Uniform Quantization (\mu-law)');  
xlabel('Time (s)');  
ylabel('Amplitude');  
grid on;
```



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